NRSG 224 — Circulatory Part 1 (Exam Notes)

High-yield only. Red = spoken detail she said that was not on the slides. Black = slides + handout.

Circulatory system overview

Three components: heart (pump), blood vessels (carry), blood (fill the vessels). Blood was already covered; this unit is heart and vessels.

Primary function: transport. What it moves:

- O2 and nutrients to cells (for energy)
- Waste away from cells to organs of elimination (lungs, kidneys)
- Electrolytes and hormones (cell function)
- Immune substances / antibodies (defense)
- Heat from core to skin (thermal regulation)

If even one of those fails (example: no O2/nutrient delivery), cells die. That is why a bad circulatory system causes so many problems.

Heart chambers and what fails when they fail

Four chambers: two atria on top, two ventricles on bottom.

Atria — two jobs:

- Collect blood returning to the heart
- Serve as pumps that fill the ventricles

If atria do not fill the ventricles → not much blood goes out per beat → cardiac output drops.

Second problem if atria fail: blood pools and stagnates in the atria → blood wants to clot. Those clots can drain into the ventricles, get pumped out, occlude arteries elsewhere, and cause ischemia and tissue damage.

Ventricles = main pumping chambers. If they fail, cardiac output drops.

Right ventricle pumps to the lungs = pulmonary circulation. Right ventricle fails → blood is not oxygenated → poor oxygenation.

Left ventricle pumps to the rest of the body = systemic circulation. Left ventricle fails → oxygenated blood does not reach the tissues → major organs can fail.

Valves: keep blood going the right direction, not backwards. Pericardium: sac that holds the heart in position in the chest and protects it. Myocardium: thick muscle layer that does the work. Endocardium: thin inner lining. (More detail next class.)

Cardiac cycle and cardiac output

Work of the heart = contract and relax, pumping blood. That cycle = cardiac cycle.

- Systole = contracts
- Diastole = relaxes and fills

Cardiac output = amount of blood pumped out of the heart per minute.

CO changes through the day with body needs. Sitting through a lecture needs little CO; playing soccer needs a lot.

CO = HR × SV

Why the equation works: number of beats per minute × amount of blood per beat = amount of blood per minute.

Faster HR → higher CO, only up to a point. If HR is too fast, the ventricles do not have time to fill between beats and CO drops. That point is roughly 150–180–200, depending on the person's fitness.

HR is mostly set by the autonomic nervous system: the balance of sympathetic vs parasympathetic activity.

Stroke volume: preload, afterload, contractility

SV = blood pumped out per beat. Determined by preload, afterload, and myocardial contractility.

Preload

- Amount of blood in the ventricle ready to be pumped out on the next beat.
- Determined by venous return. More return → more in the ventricle → more stretch of the muscle fibers → stronger contraction → bigger SV.
- Frank-Starling law (Starling's law of the heart): more blood back → more stretch → stronger force → bigger SV. Elastic analogy: more stretch, harder snap-back.
- Limit: maximum CO when fibers are stretched about 2½ times resting length. Stretch more than that and force drops — like an overstretched elastic waistband that no longer snaps back. Too much blood in the ventricles → less SV → less CO.

Afterload

- Resistance the heart must pump against to eject blood. Determined mostly by arterial blood pressure.
- Higher BP → more resistance → less blood out per beat → lower SV → lower CO.
- Bicycle-tire analogy: flat tire is easy to pump and a lot of air goes in per stroke; as pressure rises, each stroke moves less. Same for the heart ejecting into high-pressure arteries.

Cardiac contractility

- Force of ventricular contraction that is independent of preload and afterload. Due to actin–myosin interaction.
- Inotropic influences: positive increases force, negative decreases force.

- Positive inotrope example: sympathetic (fight-or-flight) stimulation. Heart rate rises and the heart pounds because force and SV also rise.
- Negative inotrope examples: hypoxia (heart muscle cannot make enough ATP for a strong contraction → weak contraction → low SV → low CO); heart attack (dead area of wall does not contract with the rest → force drops → SV and CO drop).

Why CO matters: without CO, none of the transport functions happen. Most cardiac diseases cause problems because CO drops. Interventions aim to raise CO — increase preload, decrease afterload, or increase contractility.

Blood vessels and the force that moves blood

Three kinds: arteries (heart → body), capillaries (gas and nutrient exchange), veins (body → heart).

For blood to flow you need two things: patent vessels (clear, not clogged) and a force to move the blood. That force is arterial blood pressure. If BP is too low, there is no force and circulation is poor.

Normal BP is about 120/80.

BP (mean arterial) = CO × PVR

More CO → higher BP. Phone-booth / Volkswagen-bug analogy: the more people you stuff into a fixed space, the higher the pressure. Same for blood in arteries.

PVR is set mostly by vessel diameter. Dilated vessels = more space = lower pressure. Constricted vessels = same blood in a smaller space = higher pressure.

Green items on her slide are the ones easiest to change with drugs or other interventions.

- BP too low: start an IV and give fluid (raise volume) or give a vasoconstrictor (smaller diameter → higher PVR → higher BP).
- BP too high: diuretic (lower volume), drug to slow HR (lower CO → lower BP), or vasodilator (lower PVR → lower BP).

This is why the BP-drug categories in pharmacology make sense once you know the equation.

What controls HR, volume, and vessel diameter

Three primary controllers: autonomic nervous system, kidneys, ADH.

Autonomic nervous system

- Balances sympathetic and parasympathetic to keep HR and BP in range.
- Affects HR and vessel diameter. Does not affect blood volume.
- Two centers in the medulla: vasomotor center (sympathetic effects) and cardioinhibitory center (parasympathetic effects). Together they set HR and vessel diameter.

Kidneys

- Make urine → control blood volume.
- Also run the renin-angiotensin-aldosterone system. If BP/volume drops → blood flow to the kidney drops → kidney releases renin → renin converts angiotensinogen to angiotensin I → angiotensin I is converted (in the lung) to angiotensin II → angiotensin II constricts vessels and triggers aldosterone → aldosterone holds sodium and water → volume and BP rise.

ADH (antidiuretic hormone)

- Released from the posterior pituitary when volume or BP is low.
- Vasoconstricts and holds water → volume and BP rise.

Hypertension

Primary (essential): no identifiable disease causing it. Secondary: caused by another disease (example: renal disease).

Primary is not a specific disease of one system. The three control systems (ANS, kidneys, ADH) have shifted their set point — they have been reprogrammed to hold BP at a higher level. That is not the same as a disease process; a disease process would be secondary HTN.

Guidelines were updated in 2017. Normal requires both systolic <120 and diastolic <80. If only one number is high, you are already out of normal.

Old diagnosis threshold was chronically >140/90. Under the current guidelines, 140/90 is already stage 2. The old pre-hypertension category was dropped because people heard pre and thought they did not have it yet. Stage 1 and stage 2 use or, not and: only one number has to fit the category.

Diagnose only if elevated on two or more occasions. One high reading is not enough.

Silent killer: no symptoms other than the high number itself. Occasionally headaches or nosebleeds if the pressure is extremely high, but most people feel nothing.

We care because HTN predisposes to atherosclerosis, which causes heart disease and stroke.

Two ways HTN promotes atherosclerosis:

- Mechanical stress → arterial walls hypertrophy (thicken) to withstand the pressure → lumen narrows and the wall becomes less elastic (hardened).
- Mechanical stress damages the endothelial lining.

Predisposing factors to HTN

Non-modifiable

- Family history
- Age (older; but HTN is appearing younger because modifiable risks are rising in the population)
- Race: highest in Black / African American patients — more frequent, more severe, and at a younger age. Reason is unclear; possible salt sensitivity or higher sympathetic activity.

- Sex: men higher risk until menopause; after menopause women equal men. She keeps sex in non-modifiable.

Modifiable

- Diet: high sodium (holds water → volume up); low potassium
- Obesity, especially upper-body / apple shape
- Sedentary
- Excess alcohol: more than 2 drinks/day for men, more than 1 for women. You cannot bank the week's drinks for the weekend — that is binge drinking, which also raises cancer, accident, and violence risk.
- Stress: acute stress raises BP through sympathetic vasoconstriction, but primary HTN is chronic. Current evidence points to a social component (poverty, abuse, racism, sexism). Assess anxiety, depression, and stress.
- Smoking, high cholesterol, high fat are in parentheses because they associate more strongly with heart disease than with HTN, but they matter because HTN raises heart-disease risk.

Endothelium and atherosclerosis

Endothelium is the smooth inner lining of arteries. It used to be thought of as only a slippery surface. It actually produces substances that control constriction, dilation, and platelet aggregation.

Injury to endothelium → two consequences:

- Macrophages stick to the injured area, ingest LDL, become foam cells, squeeze between endothelial cells, migrate under the lining, and die there. Their material stays in the wall.
- Endothelium changes what it releases: less antithrombotic and vasodilatory substance, more growth factors → smooth muscle proliferates → fibrous cap forms over the plaque.

Once the plaque grows, three problems:

- It takes up space and narrows the lumen → less flow.
- The fibrous cap can rupture → clot forms on the rough surface → sudden occlusion.
- Pieces of plaque or clot can embolize downstream.

DO NOT MIX UP: Arteriosclerosis = general hardening/thickening of arteries. Atherosclerosis = the specific plaque form with lipid core and fibrous cap.

Predisposing factors to atherosclerosis

Non-modifiable: family history, age, sex.

Modifiable: smoking, HTN, diabetes mellitus, hypercholesterolemia.

Also associated (more with HTN, but assess them here too): upper-body obesity, sedentary, stress, poor sleep.

Cholesterol

- LDL carries cholesterol to tissues → available for plaque. Want low.
- HDL clears cholesterol from blood and returns it to the liver → less available for plaque. Want high.

- Guidelines: total <200 (experts now lean toward <170 or even <150 in high-risk patients); HDL >40 (below 40 = high CHD risk; ≥60 is considered cardioprotective and some experts let you subtract a risk factor); LDL <100 (some aim <70 in very high risk, but that usually needs medication and the side-effect trade-off has to be weighed); triglycerides <200 (some aim <150).

Ischemia vs infarction and collateral circulation

Ischemia = not enough arterial blood flow / oxygen, but no permanent damage yet. Infarction = ischemic necrosis; the tissue is dead.

Collateral circulation: alternate small vessels that can widen and supply an area if the main artery is blocked. Present in heart, brain, and large muscle groups (legs).

Collaterals develop more easily when the blockage is gradual (atherosclerosis). That is one reason atherosclerosis is often silent until plaques are large. A sudden clot leaves no time for collaterals to open, so an acute problem is more likely.

Interventions: lifestyle modification first (stop smoking, control diabetes, control HTN, lower cholesterol). Medications if lifestyle is not enough. Goal is to taper medications as lifestyle changes take hold.

Coronary heart disease / coronary artery disease

Atherosclerosis in the coronary arteries. She prefers CAD because it names the location: impaired flow through the coronary arteries.

Heart muscle does not get its oxygen from the blood inside the chambers. It gets oxygen from blood delivered through the coronary arteries. If those arteries fail, heart muscle is damaged and function drops.

Most common cause is atherosclerosis, so charts sometimes say atherosclerotic heart disease (ASHD). Predisposing factors are the same as for atherosclerosis.

Three ways patients present:

- Asymptomatic — plaques present, no symptoms yet. Least often seen, because they have no reason to come in.
- Chronic ischemic heart disease: stable angina, variant angina
- Acute coronary syndrome: unstable angina, acute myocardial infarction

Angina types

Stable (exertional / classic / typical)

- Triggered by increased demand (exertion, emotion, cold, large meal).
- Relieved by rest or nitroglycerin.
- Pattern is predictable.

Variant (vasospastic / Prinzmetal's)

- Coronary artery spasm, not a fixed plaque.
- Can occur at rest, often at night or early morning.
- Not reliably related to exertion.

Unstable (preinfarction / crescendo)

- Plaque is unstable. Pain at rest or with minimal exertion, longer, more frequent, or new.
- Not reliably relieved by rest or NTG. Warning that infarction may be next.

Levine's sign: clenched fist over the chest. Classic pain is substernal and may radiate to the left shoulder and down the inside of the left arm.

TEST TIP: Stable = predictable, demand-related, relieved by rest/NTG. Unstable = rest or minimal activity, not relieved, progressing. Variant = spasm, often at rest.

Acute myocardial infarction

Heart muscle cell injury or death → three consequences:

- Decreased cardiac output
- Inflammation → fever, elevated WBC
- Elevated cardiac markers (CK, LD, myoglobin, troponin)

Chest pain is usually more severe than angina and often needs opioids.

Nitroglycerin

White pill in a brown bottle, under the tongue. Vasodilator.

- Dilates arteries → lower BP → lower afterload → less workload
- Dilates veins → less venous return → lower preload → less workload

Bottom line: it works by decreasing demand, not by increasing supply.

3 × 5 rule: up to 3 doses, 5 minutes apart. If pain is still there after 3, treat as possible MI.

Revascularization and other treatments

PTCA (percutaneous transluminal coronary angioplasty): balloon opens the narrowed segment.

CABG: bypass the blocked segment, often with 3–4 grafts.

Low-dose aspirin: baby aspirin daily or adult aspirin every other day. Antiplatelet. Used to prevent a first or second MI. Avoid if gastric ulcers or aneurysms (bleeding risk).

Only for MI — thrombolytic / fibrinolytic therapy:

- Not the same as an anticoagulant (heparin, Coumadin). Anticoagulants keep existing clots from growing and prevent new clots; they cannot break a clot. A thrombolytic breaks the clot.

- Goal: reopen the artery and save cells that are injured but not yet dead. Dead cells cannot be saved.
- Start within 12 hours of symptom onset; maximum benefit within 2 hours.
- Not for patients at high risk of bleeding complications.

Only for MI — opioid analgesia (morphine):

- MI pain is severe enough to need opioids.
- Severe pain drives sympathetic stimulation → tachycardia → higher heart workload → worse supply-versus-demand mismatch. Relieving the pain removes that extra demand.

TEST TIP: If she asks why a drug is used, map it to the equation: does it change HR, preload, afterload, contractility, volume, or vessel diameter? That is how she framed the interventions.